

Subnets

Segmenting a VPC into Addressable, AZ-Scoped Networks

Interview Preparation Guide

Problem & motivation - CIDR math & reserved IPs - public/private/isolated - routing & packet flow - failure modes - scaling - security - observability & automation - cost - troubleshooting - real-world patterns

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Executive Summary

A subnet is a contiguous slice of a VPC's IP address space that is bound to a single Availability Zone. Subnets are the unit at which you attach a route table and a network ACL, and they are where elastic network interfaces (and therefore instances, containers, load balancers, and databases) actually live. Whether a subnet is 'public', 'private', or 'isolated' is decided entirely by its route table and the gateways it can reach - not by any single attribute. This guide covers CIDR planning, the AZ relationship, packet flow, failure modes, scaling, security, observability, automation, cost, and troubleshooting.

1. The Problem It Solves

A flat /16 network with thousands of hosts is hard to secure, hard to reason about, and ties everything to one failure domain. Subnets let you segment that space so different tiers get different reachability, blast radius is contained, and resources can be spread across AZs.

Core needs subnets address

- **Segmentation:** Separate web/app/data tiers with distinct routing and filtering so the database is never directly internet-reachable.
- **AZ placement:** A subnet maps to exactly one AZ; spreading subnets across AZs is how you build highly available, multi-AZ architectures.
- **Blast-radius control:** Per-subnet route tables and NACLs limit how far a misconfiguration or compromise can spread.
- **Address organization:** Predictable CIDR allocation makes routing, peering, and on-prem integration tractable.

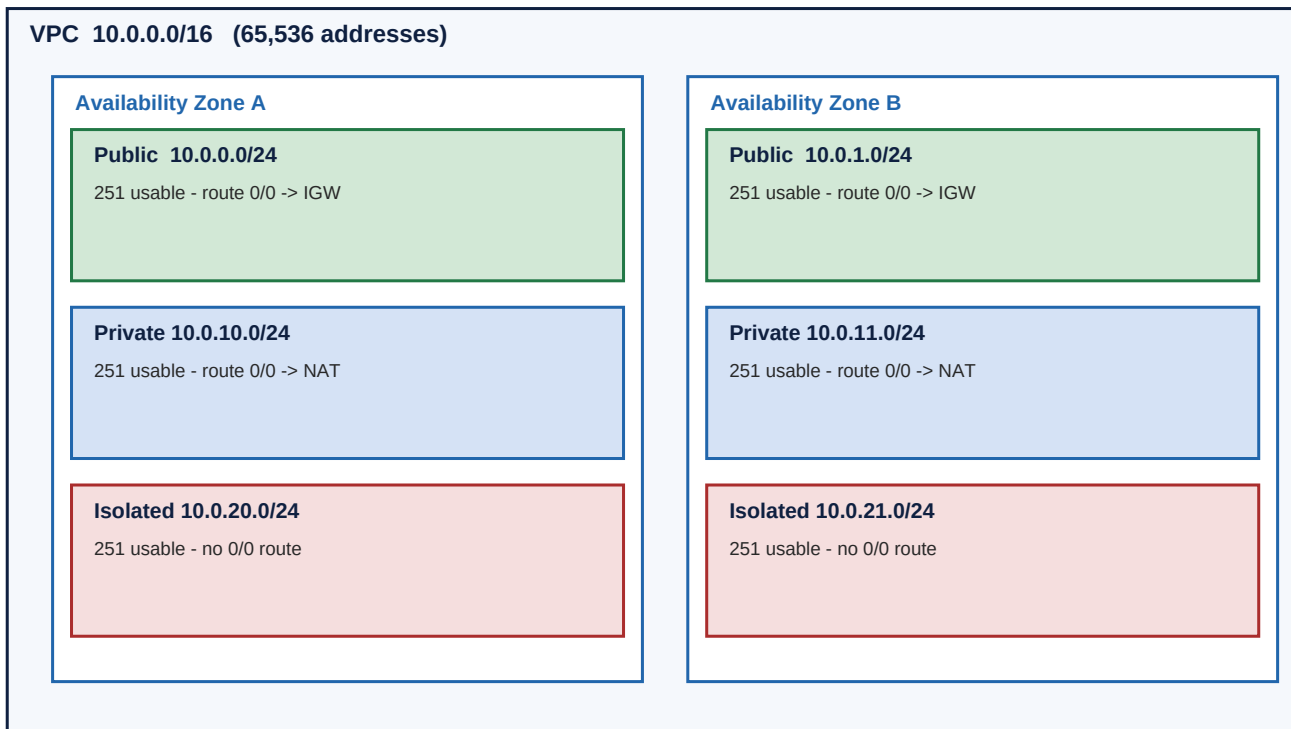
2. How It Works Internally

A subnet is defined by a CIDR block that is a subset of the VPC CIDR and must not overlap any sibling subnet. AWS reserves 5 addresses in every subnet: the network address, the implied VPC router (.1), the DNS resolver (.2), a reserved address (.3), and the broadcast address. So a /24 yields 251 usable IPs, not 256.

- Each subnet is associated with exactly one route table (the VPC 'main' table by default) and one network ACL.
- Instances launched into a subnet receive an ENI with a private IP from the subnet's range; the ENI is the real network identity.
- The subnet's AZ is fixed at creation and cannot be changed - to move AZs you create a new subnet.
- IPv6 subnets are always /64; IPv4 subnet sizing is your choice within the VPC block.

Reference Diagram - Carving a VPC into subnets

A single VPC CIDR (10.0.0.0/16, ~65k addresses) is divided into smaller per-AZ subnets. Each subnet lives in exactly one Availability Zone and is tagged public, private, or isolated by how its route table is configured.



AWS reserves 5 addresses per subnet (network, VPC router, DNS, future use, broadcast). Subnet size is fixed at creation - plan CIDRs up front. 'Public vs private' is not a flag; it is determined by the route table.

3. Traffic Flow, Packet Flow & Routing Decisions

How a subnet decides where a packet goes

- An instance in subnet 10.0.10.0/24 sends a packet; the source ENI's security group (egress) is evaluated first (stateful).
- The subnet's route table is consulted using longest-prefix match: the VPC local route covers in-VPC traffic; 0.0.0.0/0 sends the rest to IGW/NAT/etc.
- The subnet's NACL is applied at the subnet boundary (stateless) on the way out and back in.
- Return traffic is allowed automatically by the security group; the NACL must explicitly allow ephemeral ports.

Public vs private - the real definition

A subnet is 'public' only if its route table has a 0.0.0.0/0 route to an Internet Gateway AND the instance has a public/elastic IP. Remove that route and the same subnet becomes private. There is no 'isPublic' switch.

4. Advantages & Disadvantages

Advantages

- Fine-grained segmentation and per-tier reachability.
- Natural mapping to AZs for high availability.
- Smaller blast radius via per-subnet route tables and NACLs.
- Clear, auditable structure for security and compliance.

Disadvantages / costs

- CIDR sizing is permanent - undersizing forces painful re-architecture.
- More subnets means more route tables/NACLs to manage and keep consistent.
- Each AZ needs its own subnet, multiplying objects in large designs.
- IP exhaustion within a subnet silently blocks new launches.

5. Common Failure Modes

Failure	Symptom	Typical root cause
IP exhaustion	Launches fail / no free IP	Subnet CIDR too small for ENIs (esp. EKS/Fargate density).
Wrong route table	Unexpected public/no egress	Subnet associated with main table instead of intended custom table.
Single-AZ design	Outage when one AZ fails	Critical subnet not replicated across AZs.
CIDR overlap	Cannot create / peer	New subnet overlaps a sibling or peered VPC range.
NACL blocks returns	One-way connectivity	Stateless NACL missing ephemeral-port allow.
Reserved-IP confusion	Off-by-5 capacity errors	Forgetting AWS reserves 5 IPs per subnet.

6. Scaling Considerations

- **Right-size CIDRs:** Plan for peak ENI count (pods, tasks, instances), not current usage; prefer larger subnets for container platforms.
- **AZ fan-out:** Create symmetric subnets in 2-3 AZs per tier for HA.
- **Secondary CIDRs:** Attach additional CIDR blocks to a VPC when the original space runs out.
- **IPAM:** Use centralized IP Address Management to avoid overlaps across many accounts/regions.
- **Prefix delegation:** On EKS, prefix delegation packs more pods per node without exhausting subnet IPs.

7. Security Implications

- Put only internet-facing resources (ALB, NAT) in public subnets.
- Keep app tiers in private subnets and data tiers in isolated subnets with no egress route.
- Attach restrictive NACLs as a coarse, subnet-wide backstop to security groups.
- Never rely on subnet placement alone - combine with security groups and app-layer auth (zero trust).

8. Observability, Reliability & Automation

Observability

- Enable VPC Flow Logs per subnet to see accepted/rejected flows.
- Track free-IP / ENI counts to catch exhaustion before it bites.

Reliability

- Replicate every tier's subnet across multiple AZs.
- Ensure NAT/egress exists per-AZ so an AZ loss does not break others.

Automation

- Define subnets in IaC (Terraform/CloudFormation/CDK) with computed CIDRs.
- Use IPAM + policy checks to prevent overlaps and enforce naming/tags.

9. Cost Considerations

Cost driver	Why it adds up	Mitigation
Cross-AZ traffic	Per-GB charge between subnets in different AZs.	Co-locate chatty components; AZ-aware routing.
NAT per AZ	One NAT GW per AZ multiplies hourly + data cost.	Balance HA vs cost; consolidate where risk allows.
Wasted address space	Oversized subnets reserve IPs unused.	Right-size; reclaim with secondary CIDRs as needed.

10. Troubleshooting Approach

Walk the subnet's configuration in order:

- 1. Which subnet/AZ is the ENI in, and is there a free IP?
- 2. Which route table is associated, and what is the 0.0.0.0/0 target?
- 3. Do the NACL rules allow both directions (incl. ephemeral ports)?
- 4. Do security groups on the ENI allow the flow?
- 5. Read Flow Logs and run Reachability Analyzer for proof.

Mnemonic

'AZ -> Route table -> NACL -> SG -> Flow logs.' Most subnet connectivity issues are a wrong route-table association or an ephemeral-port NACL gap.

11. Real-World Examples & Patterns

- **3-tier per AZ:** public (ALB/NAT) + private (app) + isolated (DB) subnets replicated across 2-3 AZs.
- **EKS large subnets:** Use /20 or larger private subnets to accommodate per-pod ENIs and prefix delegation.
- **Shared egress:** Small public subnets hosting only NAT; large private subnets for workloads.
- **Landing zone:** IPAM-managed, non-overlapping subnet CIDRs standardized across many accounts.

12. Senior-Level Trade-offs & Talking Points

- Few large subnets vs many small subnets: simpler IP management vs finer isolation and blast-radius control.
- Subnet-per-tier-per-AZ object explosion vs operational simplicity.
- Reserve generous CIDRs now (cheap) vs re-architecting later (expensive).
- NACL backstop vs complexity: stateless rules add safety but are easy to misorder.

Whiteboard checklist

(1) Draw VPC CIDR. (2) Slice per-AZ public/private/isolated subnets. (3) Show route-table association per subnet. (4) Mark reserved IPs and usable count. (5) Trace a packet out and back. (6) Discuss AZ HA, IP exhaustion, and cross-AZ cost.

13. Quick Glossary

Term	Meaning
CIDR	IP range notation (e.g., /24 = 256 addresses, 251 usable).
AZ	Availability Zone; a subnet belongs to exactly one.
ENI	Elastic Network Interface - the real network identity of a resource.
Public subnet	Subnet whose route table sends 0/0 to an Internet Gateway.
Isolated subnet	Subnet with no default route off the VPC.
Secondary CIDR	Extra address block added to a VPC when space runs out.
IPAM	IP Address Management for planning/allocating non-overlapping CIDRs.